

**LAPORAN UJIAN AKHIR SEMESTER
PRAKTIKUM JARINGAN KOMPUTER
TOPOLOGI HYBRID DALAM WARNET MEDAN EMPIRE 3 TIER**



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**PROGRAM STUDI S1 TEKNIK INFORMATIKA
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1. Latar Belakang

Seiring dengan perkembangan teknologi informasi yang pesat, akses terhadap jaringan komputer dan internet telah menjadi kebutuhan utama bagi masyarakat, baik untuk keperluan pendidikan, hiburan, komunikasi, maupun bisnis. Salah satu bentuk layanan publik yang menyediakan akses tersebut adalah warung internet (warnet).

Namun, dalam praktiknya, masih banyak warnet yang memiliki jaringan tidak terstruktur dengan baik, menyebabkan koneksi internet sering bermasalah, kecepatan akses tidak merata antar komputer, hingga kesulitan dalam proses pemeliharaan jaringan. Kondisi ini tentu mengurangi kualitas layanan yang diberikan kepada pengguna.

Masalah utama yang kami angkat dalam proyek ini adalah kurangnya perancangan topologi jaringan yang efisien dan handal pada sistem warnet. Masalah ini berdampak pada kenyamanan pengguna, meningkatnya biaya pemeliharaan, serta potensi kerugian bagi pemilik warnet.

Kami memilih masalah ini karena jaringan komputer merupakan bagian terpenting dari operasional sebuah warnet. Dengan merancang topologi jaringan yang tepat, kami dapat memberikan solusi nyata yang tidak hanya meningkatkan efisiensi sistem, tetapi juga dapat diterapkan langsung di lapangan.

Secara teori, jaringan komputer adalah sekumpulan komputer dan perangkat lainnya yang saling terhubung untuk berbagi sumber daya dan informasi. Dalam perancangannya, dikenal berbagai jenis topologi seperti star, bus, ring, mesh, dan hybrid, yang masing-masing memiliki kelebihan dan kekurangannya. Pemilihan jenis topologi yang sesuai sangat penting untuk menjamin kinerja dan stabilitas jaringan.

Solusi yang kami tawarkan adalah membangun sebuah topologi jaringan warnet yang disesuaikan dengan kebutuhan operasional, jumlah perangkat, dan skala layanan. Desain ini mencakup pemilihan perangkat keras (seperti switch, router, dan kabel), pemetaan alur koneksi antar komputer, hingga pertimbangan aspek keamanan dan kemudahan pemeliharaan.

Dengan proyek ini, kami berharap dapat memberikan kontribusi dalam bentuk perancangan jaringan yang praktis, efisien, dan mudah diterapkan untuk mendukung pelayanan internet yang lebih baik di lingkungan warnet.

2. Tujuan dan Manfaat

a. Tujuan :

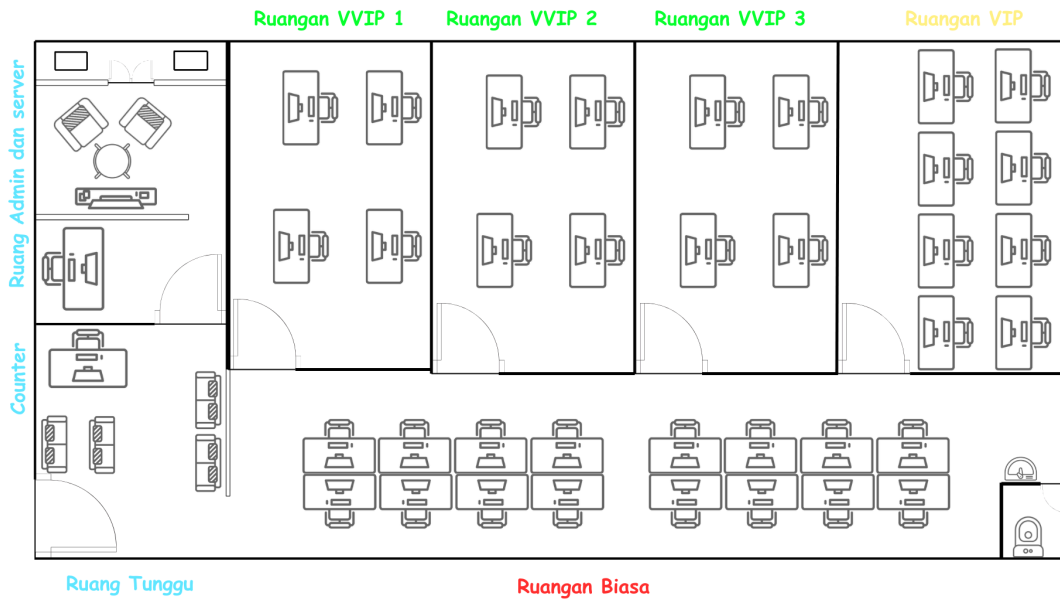
- Merancang topologi jaringan yang sesuai untuk kebutuhan operasional sebuah warnet.
- Menentukan perangkat keras dan lunak yang dibutuhkan dalam pembangunan jaringan.
- Meningkatkan efisiensi dan kestabilan koneksi antar komputer dalam lingkungan warnet.
- Menyusun sistem manajemen jaringan yang mendukung keamanan dan kemudahan pemeliharaan.
- Memberikan solusi desain jaringan yang dapat diimplementasikan secara nyata dan ekonomis.

b. Manfaat :

- Bagi pemilik warnet: Mendapatkan rancangan sistem jaringan yang efisien dan siap digunakan, sehingga dapat menekan biaya operasional dan memaksimalkan keuntungan.
- Bagi teknisi jaringan: Menjadi acuan dalam pemasangan, pengelolaan, dan pemeliharaan sistem jaringan warnet secara efektif.
- Bagi pengguna/pelanggan warnet: Menikmati layanan internet yang cepat, stabil, dan minim gangguan.
- Bagi akademisi dan pelajar: Menjadi bahan studi atau referensi dalam memahami perancangan dan implementasi jaringan komputer.
- Secara umum: Mendukung peningkatan literasi digital masyarakat dengan menyediakan akses internet yang handal melalui infrastruktur warnet.

3. Skema dan Deskripsi Lokasi Kasus

a. Skema Lokasi

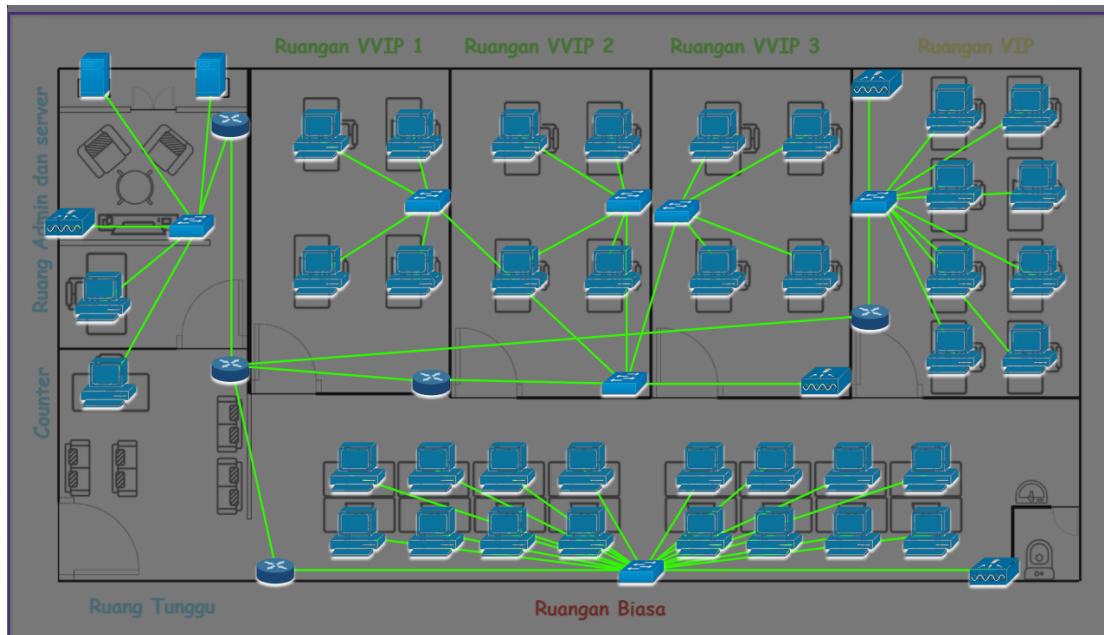


b. Deskripsi Lokasi

Warnet Medan terdiri dari 1 lantai yang berisi 7 ruangan. Pada saat masuk warnet kita langsung masuk ke dalam ruang tunggu dan counter yang berisi 1 komputer dan 1 router internet. Pada sebelah kiri ruang tunggu dan counter terdapat ruang admin dan server yang berisi 1 komputer, 1 access point, 1 router, 1 switch, dan 2 server (DNS Server dan Web Server). Setelah ruangan tunggu kita akan melewati ruangan pada tingkat 1 atau ruangan biasa yang berisi 1 router, 1 access point, 1 switch, dan 16 komputer. Setelah ruangan tingkat 1, terdapat 4 ruangan VVIP 1, 2, 3 yang merupakan ruangan tingkat 3, dan ruangan VIP yang merupakan ruangan tingkat 2. Ruangan tingkat 2 atau VIP berisi 1 router, 1 switch, 1 access point, dan 8 komputer. Ruangan tingkat 3 atau VVP berisi 1 router, 1 access point, 4 switch (1 switch utama, dan 3 switch cabang untuk setiap ruangan), 4 komputer setiap ruangan, sehingga terdapat 16 komputer untuk 3 ruangan VVIP.

4. Deskripsi Rancangan Jaringan

a. Skema Topologi yang Akan Digunakan



b. Deskripsi Topologi dan Perangkat yang Akan Digunakan

Topologi yang digunakan:

Kami menggunakan topologi hybrid yang merupakan gabungan dari beberapa jenis topologi, yang terdiri atas:

a) Topologi Star

Topologi ini digunakan pada zona lokal ruangan tingkat 1 atau ruangan biasa dengan perangkat 16 PC, 1 access point, 1 switch dan 1 router. Koneksi yang terjadi antara perangkat tersebut, yaitu semua PC dan access point akan terhubung ke switch, switch terhubung ke router, dan router terhubung ke router internet. Topologi ini berfungsi untuk melayani pengguna umum dari warnet. Topologi ini memudahkan untuk manajemen jaringan jika terjadi masalah dan skalabilitasnya tinggi sehingga penambahan perangkat lebih mudah dengan menggunakan port dari switch.

Topologi ini juga digunakan pada zona lokal ruangan tingkat 2 atau ruangan VIP dengan perangkat yang digunakan, yaitu 8 PC, 1 access point, 1 switch, dan 1 router. Koneksi yang terjadi antara semua PC dan access point ke switch kemudian ke router. Topologi ini berfungsi untuk melayani ruang tambahan atau sebagai cadangan bagi

pengguna jika ruangan biasa atau tingkat 1 penuh. Topologi ini dapat menambah kapasitas tanpa mengganggu zona utama yang telah ada.

Topologi ini juga digunakan pada ruangan Admin dan Server dengan perangkat 1 PC, 1 DNS Server, 1 Web Server, 1 Access Point, 1 switch dan 1 router. Koneksi yang terjadi, yaitu semua perangkat terhubung ke switch, kemudian switch akan terhubung ke router. Ruangan ini berfungsi sebagai server penyedia layanan DNS dan Web dan memberikan akses administratif jaringan. Topologi ini memberikan kontrol yang terpusat

b) Topologi Tree

Topologi ini digunakan pada ruangan tingkat 3 atau ruangan VVIP dengan perangkat 4 PC setiap ruangan, 4 switch (1 switch utama dan 3 switch cabang), 1 access point, dan 1 router. Koneksi yang terjadi antar router ke switch utama, kemudian ke 3 switch cabang yang masing-masing cabang akan terkoneksi dengan 4 PC. Topologi ini berfungsi untuk melayani pengguna VIP dengan kecepatan tinggi, segmentasi VLAN, dan keamanan yang lebih baik. Topologi ini memiliki segmentasi yang lebih baik karena dibagi menjadi beberapa cabang, sehingga data antar kelompok tidak saling terganggu dan bandwidth khusus karena dibagi percabang, sehingga koneksi akan lebih stabil dan cepat.

c) Topologi Point-to-Point

Topologi ini digunakan pada internet yang menghubungkan router pusat dari setiap ruangan ke router internet sehingga berfungsi sebagai penyedia rute menuju ke internet dengan titik distribusi lalu lintas. Topologi ini berfungsi sebagai penghubung antar zona. Topologi ini membuat routing lebih efisien dan mempercepat distribusi.

Perangkat:

Router 0 → digunakan sebagai Router utama yang menghubungkan 4 ruangan, yaitu ruangan departemen Regular, VIP, VVIP, dan Server.

Router 1 → digunakan sebagai Router Local Area Network ruang Regular

Router 2 → digunakan sebagai Router Local Area Network ruang VIP

Router 3 → digunakan sebagai Router Local Area Network ruang VVIP

Router 4 → digunakan sebagai Router Local Area Network ruang Server

Switch 1 → Switch utama pada Local Area Network ruang Regular

Switch 2 → Switch utama pada Local Area Network ruang VIP

Switch 3 → Switch utama pada Local Area Network ruang VVIP

Switch 3.1 → Switch cabang 1 pada Local Area Network ruang VVIP

Switch 3.2 → Switch cabang 2 pada Local Area Network ruang VVIP

Switch 3.3 → Switch cabang 3 pada Local Area Network ruang VVIP

Switch 4 → Switch utama pada Local Area Network ruang Server

Access Point 1 → untuk WLAN ruang Regular

Access Point 2 → untuk WLAN ruang VIP

Access Point 3 → untuk WLAN ruang VVIP

Access Point 4 → untuk WLAN ruang Server

Server Web → Server untuk Web Warnet

Server DNS → Server untuk DNS website Warnet

5. Subnetting Table

Subnet Name	Host Count	Network	Gateway	Host Range	Broadcast
Regular	16	192.168.10.0 / 24	192.168.10.1	192.168.10.100 - 192.168.10.115	192.168.10.116
VIP	8	192.168.20.0 / 24	192.168.20.1	192.168.20.100 - 192.168.20.107	192.168.20.108
VVIP	12	192.168.30.0 / 24	192.168.30.1	192.168.30.100 - 192.168.30.111	192.168.30.111
Server	2	192.168.99.0 / 24	192.168.99.1	192.168.99.100 - 192.168.99.101	192.168.99.102

6. Addressing Table

Device Name	Interface / Port	IP Address / Prefix	Address Pool	Keterangan
-------------	------------------	---------------------	--------------	------------

R0	F0/3/0	10.0.1.1/30	STATIC	IPv4 Link to R1
	F0/3/1	10.0.2.1/30	STATIC	IPv4 Link to R2
	F0/3/2	10.0.3.1/30	STATIC	IPv4 Link to R3
	F0/3/3	10.0.4.1/30	STATIC	IPv4 Link to R4
R1	G0/0/0	192.168.10.1 / 24	192.168.10.100 - 192.168.10.115	IPv4 Link to Regular Room.
		2001:DB8:10::1/64	2001:DB8:10:: (STATELESS)	IPv6 Link to Regular Room.
	G0/0/1	10.0.1.2/ 24	STATIC	IPv4 Link to R0.
				IPv6 Link to R0.
R2	G0/0/0	192.168.20.1 / 24	192.168.20.100 - 192.168.20.107	IPv4 Link to VIP Room.
		2001:DB8:20::1/64	2001:DB8:10:: (STATELESS)	IPv6 Link to VIP Room.
	G0/0/1	10.0.2.2/ 24	STATIC	IPv4 Link to R0.
				IPv6 Link to R0.
R3	G0/0/0	192.168.30.1 / 24	192.168.30.100 - 192.168.30.111	IPv4 Link to VVIP Room.
		2001:DB8:30::1/64	2001:DB8:30:: (STATELESS)	IPv6 Link to VVIP Room.
	G0/0/1	10.0.3.2/ 24	STATIC	IPv4 Link to R0.
				IPv6 Link to R0.
R4	G0/0/0	192.168.99.1/ 24	192.168.30.100 - 192.168.30.111	IPv4 Link to Server Room.
		2001:DB8:D::2/64	2001:DB8:30::	IPv6 Link to Server Room.

			(STATELESS)	
	G0/0/1	10.0.4.2/ 24	STATIC	IPv4 Link to R0.
				IPv6 Link to R0.
PC Regular Room	NIC	DHCPv4 Client	DHCPv4 Client	IPv4 Client from R1 G0/0/0
		DHCPv6 Client	DHCPv6 Client	IPv6 Client from R1 G0/0/0
PC VIP Room	NIC	DHCPv4 Client	DHCPv4 Client	IPv4 Client from R2 G0/0/0
		DHCPv6 Client	DHCPv6 Client	IPv6 Client from R2 G0/0/0
PC VVIP Room	NIC	DHCPv4 Client	DHCPv4 Client	IPv4 Client from R3 G0/0/0
		DHCPv6 Client	DHCPv6 Client	IPv6 Client from R3 G0/0/0
PC Server Room	NIC	DHCPv4 Client	DHCPv4 Client	IPv4 Client from R4 G0/0/0
		DHCPv6 Client	DHCPv6 Client	IPv6 Client from R4 G0/0/0
DNS Server	NIC	192.168.99.11/24	STATIC	DNS Server Warnet
		2001:DB8:99::11/64	STATIC	DNS Server Warnet
Web Server	NIC	192.168.99.10/24	STATIC	Web Server Warnet
		2001:DB8:99::10/64	STATIC	Web Server Warnet

7. Rencana Eksplorasi

- a. Virtual LAN (VLAN) [Voice, Data, Management, Trunking]
 - VLAN digunakan untuk route antara Router Pusat (R0) ke router lainnya karena kami menggunakan modul HWIC-4ESW yang tidak bisa di setting IP nya dalam router sehingga kami menggunakan VLAN
- b. Wireless LAN (WLAN)
 - WLAN digunakan agar pengguna bisa menggunakan perangkat wireless dalam warnet kami
- c. Server Web dan DNS
 - Server Web sebagai server website dari warnet itu sendiri dan Server DNS yang bekerja sebagai Domain web Warnet dimana pengguna bisa menambah waktu pemakaian PC dan bisa memesan makanan

d. Subnetting IPv6

→ make IPv6 di masing-masing LAN

8. Screenshot Command Konfigurasi di Setiap Perangkat

- Router0 (Internet)

```
R0#show r
Building configuration...

Current configuration : 2888 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R0
!
!
!
!
!
!
!
ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
username admin privilege 15 secret 5 $1$mERr$AFx/pZTlLh7NP3Dp3P/qQ/
username user secret 5 $1$mERr$pRhYieRsLJ9jhTe0lpBTz/
!
!
license udi pid CISCO2911/K9 sn FTX1524S401-
!
!
!
!
!
!
!
ip domain-name belajar.com
--More-- |
```

```
spanning-tree mode pvst
!
!
!
!
!
!
!
interface GigabitEthernet0/0
description Internet link
ip address 203.0.113.1 255.255.255.252
ip nat outside
duplex auto
speed auto
ipv6 address 2001:DB8:113::1/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/1
ip address 8.8.8.8 255.255.255.0
duplex auto
speed auto
ipv6 address 2001:DB8:8::8/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface FastEthernet0/3/0
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/3/1
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/3/2
switchport access vlan 30
```

```

interface FastEthernet0/3/2
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/3/3
 switchport access vlan 40
 switchport mode access
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan10
 description Link to R1
 mac-address 00e0.b090.2001
 ip address 10.0.1.1 255.255.255.252
 ip nat inside
 ipv6 address 2001:DB8:1::1/64
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface Vlan20
 description Link to R2
 mac-address 00e0.b090.2002
 ip address 10.0.2.1 255.255.255.252
 ip nat inside
 ipv6 address 2001:DB8:2::1/64
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface Vlan30
 description Link to R3
 mac-address 00e0.b090.2003
 ip address 10.0.3.1 255.255.255.252
 ip nat inside
 ipv6 address 2001:DB8:3::1/64
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface Vlan40
 description Link to R4

line con 0
!
line aux 0
!
line vty 0 4
 login local
 transport input ssh
!
!
!
end

```

```

interface Vlan40
 description Link to R4
 mac-address 00e0.b090.2004
 ip address 10.0.4.1 255.255.255.252
 ip nat inside
 ipv6 address 2001:DB8:4::1/64
 ipv6 enable
 ipv6 ospf 1 area 0
!
router ospf 1
 router-id 1.1.1.1
 log-adjacency-changes
 network 10.0.1.0 0.0.0.3 area 0
 network 10.0.2.0 0.0.0.3 area 0
 network 10.0.3.0 0.0.0.3 area 0
 network 10.0.4.0 0.0.0.3 area 0
!
router rip
!
ipv6 router ospf 1
 router-id 1.1.1.1
 log-adjacency-changes
!
ip nat inside source list 1 interface GigabitEthernet0/0 overload
ip classless
ip route 0.0.0.0 0.0.0.0 203.0.113.2
!
ip flow-export version 9
!
!
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
access-list 1 permit 192.168.40.0 0.0.0.255
access-list 100 permit tcp host 192.168.99.100 any eq 22
access-list 100 permit tcp host 192.168.99.101 any eq 22
access-list 100 deny tcp any any eq 22
access-list 100 permit ip any any

```

- Router 1 (Normal)

```

!
!
!
!
!
!
hostname R1
!
!
!
ip dhcp excluded-address 192.168.10.1 192.168.10.99
!
ip dhcp pool POOL-NORMAL
dns-server 192.168.99.11
ip dhcp pool POOL-R1
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 192.168.99.11
!
!
!
ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
ip dhcp pool V6POOL-R1
dns-server 2001:DB8::11
!
!
!
username admin privilege 15 secret 5 $l$mErR$AFx/pZTlLh7NF3Dp3P/qq/
username user secret 5 $l$mErR$pRhYIeRsLJ9jhTe0lpBTz/
!
!
!
license udi pid CISCO2911/K9 sn FTX1S24HZ59-
!
!
!
!
!
!
!
!
!
!
ip domain-name belajar.com

router ospf 1
router-id 2.2.2.2
log-adjacency-changes
network 10.0.1.0 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
!
ipv6 router ospf 1
router-id 2.2.2.2
log-adjacency-changes
!
ip nat inside source list 1 interface GigabitEthernet0/2 overload
ip classless
ip route 0.0.0.0 0.0.0.0 GigabitEthernet0/2
ip route 0.0.0.0 0.0.0.0 203.0.113.1
!
ip flow-export version 9
!
!
access-list 100 permit tcp host 192.168.99.100 any eq 22
access-list 100 permit tcp host 192.168.99.101 any eq 22
access-list 100 deny tcp any any eq 22
access-list 100 permit ip any any
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login local
transport input ssh
!
!
!
end

```

- Router 2 (VIP)

```

hostname R2
!
!
!
ip dhcp excluded-address 192.168.20.1 192.168.20.99
!
ip dhcp pool POOL-R2
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 192.168.99.11
!
!
!
ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
ipv6 dhcp pool V6POOL-R2
dns-server 2001:DB8:99::11
!
!
!
username admin privilege 15 secret 5 $1$mERr$AFX/pZTlLh7NP3Dp3P/qq/
username user secret 5 $1$mERr$RhYieRsLJ9jhTe0lpBTz/
!
!
!
license udi pid CISCO2911/K9 sn FTX15248V96-
!
!
!
!
!
!
ip domain-name belajar.com
,

router ospf 1
router-id 3.3.3.3
log-adjacency-changes
network 10.0.2.0 0.0.0.3 area 0
network 192.168.20.0 0.0.0.255 area 0
!
ipv6 router ospf 1
router-id 3.3.3.3
log-adjacency-changes
!
ip classless
!
ip flow-export version 9
!
!
access-list 100 permit tcp host 192.168.99.100 any eq 22
access-list 100 permit tcp host 192.168.99.101 any eq 22
access-list 100 deny tcp any any eq 22
access-list 100 permit ip any any
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login local
transport input ssh
!
!
!
end

spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
description Link to LAN
ip address 192.168.20.1 255.255.255.0
ip access-group 100 in
duplex auto
speed auto
ipv6 address 2001:DB8:20::1/64
ipv6 nd other-config-flag
ipv6 enable
ipv6 ospf 1 area 0
ipv6 dhcp server V6POOL-R2
!
interface GigabitEthernet0/1
description Link to R0
ip address 10.0.2.2 255.255.255.252
duplex auto
speed auto
ipv6 address 2001:DB8:2::2/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
router-id 3.3.3.3
log-adjacency-changes

```

- Router 3 (VVIP)

```
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
interface GigabitEthernet0/0
description Link to LAN
ip address 192.168.30.1 255.255.255.0
ip access-group 100 in
duplex auto
speed auto
ipv6 address 2001:DB8:30::1/64
ipv6 nd other-config-flag
ipv6 enable
ipv6 ospf 1 area 0
ipv6 dhcp server V6POOL-R3
!
!
interface GigabitEthernet0/1
description Link to R0
ip address 10.0.3.2 255.255.255.252
ip access-group 100 in
duplex auto
speed auto
ipv6 address 2001:DB8:3::2/64
ipv6 enable
ipv6 ospf 1 area 0
!
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
!
interface Vlan1
no ip address
shutdown
!
!
!
hostname R3
!
!
!
!
!
ip dhcp excluded-address 192.168.30.1 192.168.30.99
!
!
ip dhcp pool POOL-R3
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
dns-server 192.168.99.11
!
!
!
!
ip cef
ipv6 unicast-routing
!
!
no ipv6 cef
!
!
ip dhcp pool V6POOL-R3
dns-server 2001:DB8:99::11
!
!
!
username admin privilege 15 secret 5 $1$mERr$AFx/pZTlLh7NF3Dp3P/qQ/
username user secret 5 $1$mERr$pRhYieRsLJ9jhTe0lpBTz/
!
!
!
license udi pid CISCO2911/K9 sn FTX1524T945-
!
!
!
!
!
!
!
!
!
!
ip domain-name belajar.com

router ospf 1
router-id 4.4.4.4
log-adjacency-changes
network 10.0.3.0 0.0.0.3 area 0
network 192.168.30.0 0.0.0.255 area 0
!
!
ipv6 router ospf 1
router-id 4.4.4.4
log-adjacency-changes
!
!
!
ip classless
!
!
!
ip flow-export version 9
!
!
!
access-list 100 permit tcp host 192.168.99.100 any eq 22
access-list 100 permit tcp host 192.168.99.101 any eq 22
access-list 100 deny tcp any any eq 22
access-list 100 permit ip any any
!
!
!
!
line con 0
!
!
line aux 0
!
!
line vty 0 4
login local
transport input ssh
!
!
!
end
```

- Router 4 (Admin)

```
hostname R4
!
!
!
!
ip dhcp excluded-address 192.168.99.1 192.168.99.99
!
ip dhcp pool POOL-R4
network 192.168.99.0 255.255.255.0
default-router 192.168.99.1
dns-server 192.168.99.11
!
!
!
ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
ipv6 dhcp pool V6POOL-R4
dns-server 2001:DB8:99::11
!
!
!
username admin privilege 15 secret 5 $l$mERr$AFx/pZTlLh7NP3Dp3P/qq/
username user secret 5 $l$mERr$pRhYIeRsLJ9jhTe0lpBTz/
!
!
!
license udi pid CISCO2911/K9 sn FTX1524PDME-
!
!
!
!
!
!
!
!
!
!
ip domain-name belajar.com
```

```

router ospf 1
router-id 5.5.5.5
log-adjacency-changes
network 10.0.4.0 0.0.0.3 area 0
network 192.168.40.0 0.0.0.255 area 0
network 192.168.99.0 0.0.0.255 area 0
!
ipv6 router ospf 1
router-id 5.5.5.5
log-adjacency-changes
!
ip classless
!
ip flow-export version 9
!
!
access-list 100 permit tcp host 192.168.99.100 any eq 22
access-list 100 permit tcp host 192.168.99.101 any eq 22
access-list 100 deny tcp any any eq 22
access-list 100 permit ip any any
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login local
transport input ssh
!
!
!
end

```

```
spanning-tree mode pvst
!  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
description Link to LAN  
ip address 192.168.99.1 255.255.255.0  
ip access-group 100 in  
duplex auto  
speed auto  
ipv6 address 2001:DB8:99::1/64  
ipv6 nd other-config-flag  
ipv6 enable  
ipv6 ospf 1 area 0  
ipv6 dhcp server V6POOL-R4  
!  
interface GigabitEthernet0/1  
description Link to R0  
ip address 10.0.4.2 255.255.255.252  
duplex auto  
speed auto  
ipv6 address 2001:DB8:4::2/64  
ipv6 enable  
ipv6 ospf 1 area 0  
!  
interface GigabitEthernet0/2  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface Vlan1  
no ip address  
shutdown  
!
```

- DNS Server

DNS Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type **AAAA Record**

Address

Add **Save** **Remove**

No.	Name	Type	Detail
0	warnetmedan.com	A Record	192.168.99.10
1	warnetmedan.com	AAAA Record	2001:DB8:99::10

DNS Server

Physical Config Services **Desktop** Programming Attributes

IP Configuration [X]

IP Configuration

☐ DHCP ☒ Static

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address / 64

Link Local Address

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

Username

Password

- Web Server

Web Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

HTTP

☒ On ☐ Off

HTTPS

☒ On ☐ Off

File Manager

	File Name	Edit	Delete
1	beranda.html	(edit)	(delete)
2	copyrights.html	(edit)	(delete)
3	cscoptlogo177x111.jpg		(delete)
4	game1.html	(edit)	(delete)
5	game2.html	(edit)	(delete)
6	helloworld.html	(edit)	(delete)
7	image.html	(edit)	(delete)
8	index.html	(edit)	(delete)
9	komputer.html	(edit)	(delete)
10	logout.html	(edit)	(delete)

Web Server

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.99.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.99.1

DNS Server: 192.168.99.11

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: 2001:DB8:99::10 / 64

Link Local Address: FE80::201:C9FF:FE86:59C3

Default Gateway: 2001:DB8:99::1

DNS Server: FE80::204:9AFF:FEA7:735B

802.1X

☐ Use 802.1X Security

Authentication: MD5

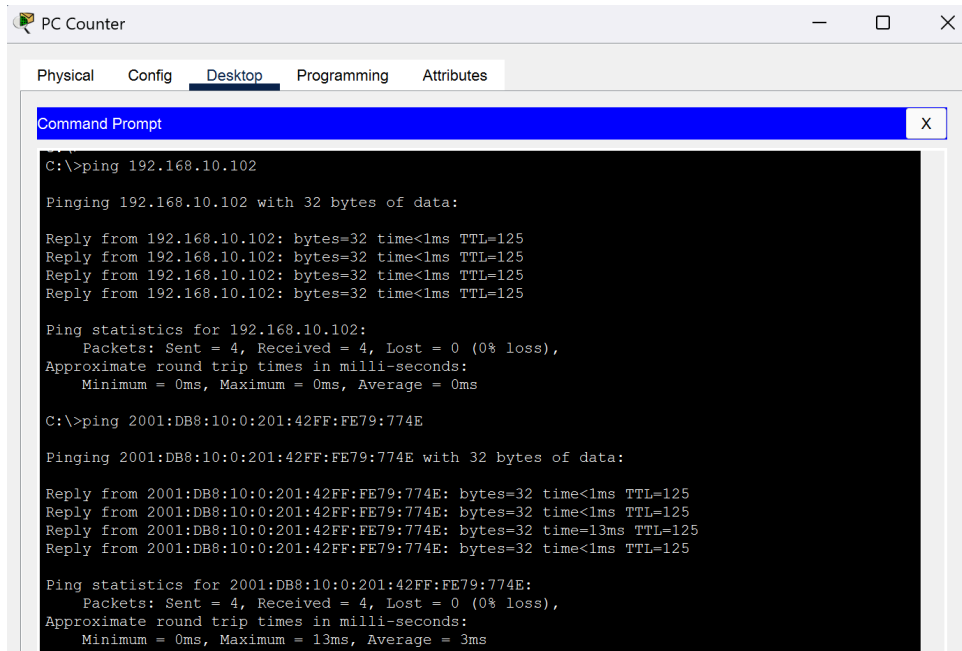
Username:

Password:

9. Cek Konektivitas

Konektivitas Perangkat Server & Admin

- LAN Server & Admin ke LAN Regular (PC-Counter ke PC 14)



```
PC Counter
Physical Config Desktop Programming Attributes

Command Prompt
C:\>ping 192.168.10.102

Pinging 192.168.10.102 with 32 bytes of data:

Reply from 192.168.10.102: bytes=32 time<1ms TTL=125
Reply from 192.168.10.102: bytes=32 time<1ms TTL=125
Reply from 192.168.10.102: bytes=32 time<1ms TTL=125
Reply from 192.168.10.102: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.10.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

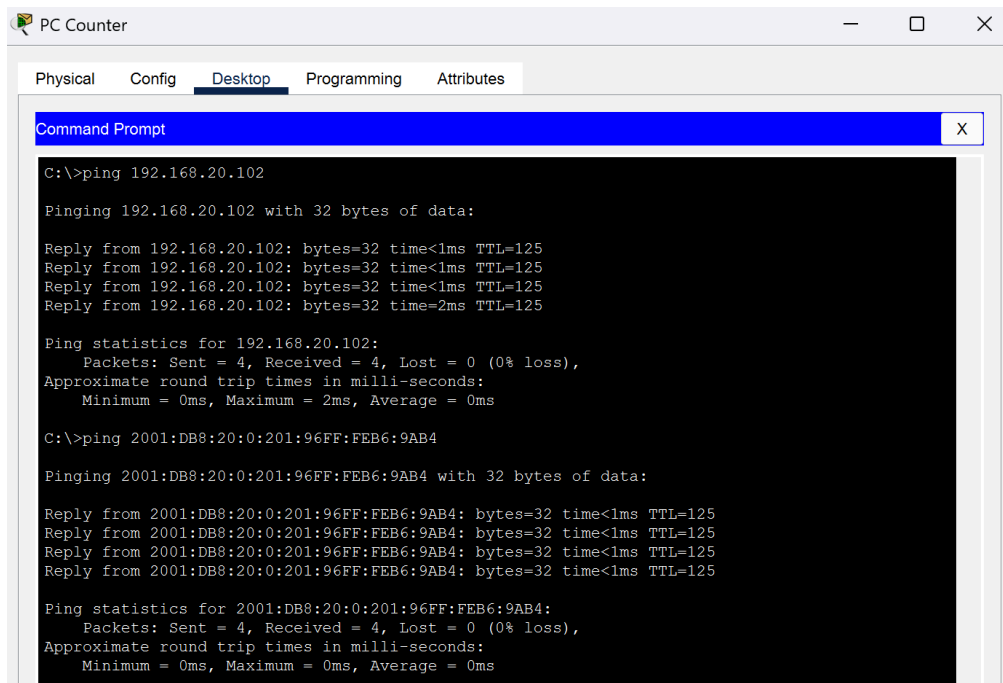
C:\>ping 2001:DB8:10:0:201:42FF:FE79:774E

Pinging 2001:DB8:10:0:201:42FF:FE79:774E with 32 bytes of data:

Reply from 2001:DB8:10:0:201:42FF:FE79:774E: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:10:0:201:42FF:FE79:774E: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:10:0:201:42FF:FE79:774E: bytes=32 time=13ms TTL=125
Reply from 2001:DB8:10:0:201:42FF:FE79:774E: bytes=32 time<1ms TTL=125

Ping statistics for 2001:DB8:10:0:201:42FF:FE79:774E:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms
```

- LAN Server & Admin ke LAN VIP (PC Counter ke PC 17(5))



```
PC Counter
Physical Config Desktop Programming Attributes

Command Prompt
C:\>ping 192.168.20.102

Pinging 192.168.20.102 with 32 bytes of data:

Reply from 192.168.20.102: bytes=32 time<1ms TTL=125
Reply from 192.168.20.102: bytes=32 time<1ms TTL=125
Reply from 192.168.20.102: bytes=32 time<1ms TTL=125
Reply from 192.168.20.102: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.20.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 2001:DB8:20:0:201:96FF:FEB6:9AB4

Pinging 2001:DB8:20:0:201:96FF:FEB6:9AB4 with 32 bytes of data:

Reply from 2001:DB8:20:0:201:96FF:FEB6:9AB4: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:20:0:201:96FF:FEB6:9AB4: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:20:0:201:96FF:FEB6:9AB4: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:20:0:201:96FF:FEB6:9AB4: bytes=32 time<1ms TTL=125

Ping statistics for 2001:DB8:20:0:201:96FF:FEB6:9AB4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- LAN Server & Admin ke PC VVIP (PC Counter ke PC16(6))

PC Counter

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>ping 192.168.30.100

Pinging 192.168.30.100 with 32 bytes of data:

Reply from 192.168.30.100: bytes=32 time<1ms TTL=125
Reply from 192.168.30.100: bytes=32 time<1ms TTL=125
Reply from 192.168.30.100: bytes=32 time<1ms TTL=125
Reply from 192.168.30.100: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.30.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 001:DB8:30:0:20B:BEFF:FED2:BE9C

Pinging 001:DB8:30:0:20B:BEFF:FED2:BE9C with 32 bytes of data:

Reply from 001:DB8:30:0:20B:BEFF:FED2:BE9C: bytes=32 time<1ms TTL=125
Reply from 001:DB8:30:0:20B:BEFF:FED2:BE9C: bytes=32 time<1ms TTL=125
Reply from 001:DB8:30:0:20B:BEFF:FED2:BE9C: bytes=32 time<1ms TTL=125
Reply from 001:DB8:30:0:20B:BEFF:FED2:BE9C: bytes=32 time<1ms TTL=125

Ping statistics for 001:DB8:30:0:20B:BEFF:FED2:BE9C:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

- LAN Server & Admin ke Server Web

PC Counter

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>ping 192.168.99.10

Pinging 192.168.99.10 with 32 bytes of data:

Reply from 192.168.99.10: bytes=32 time<1ms TTL=128
Reply from 192.168.99.10: bytes=32 time<1ms TTL=128
Reply from 192.168.99.10: bytes=32 time<1ms TTL=128
Reply from 192.168.99.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.99.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

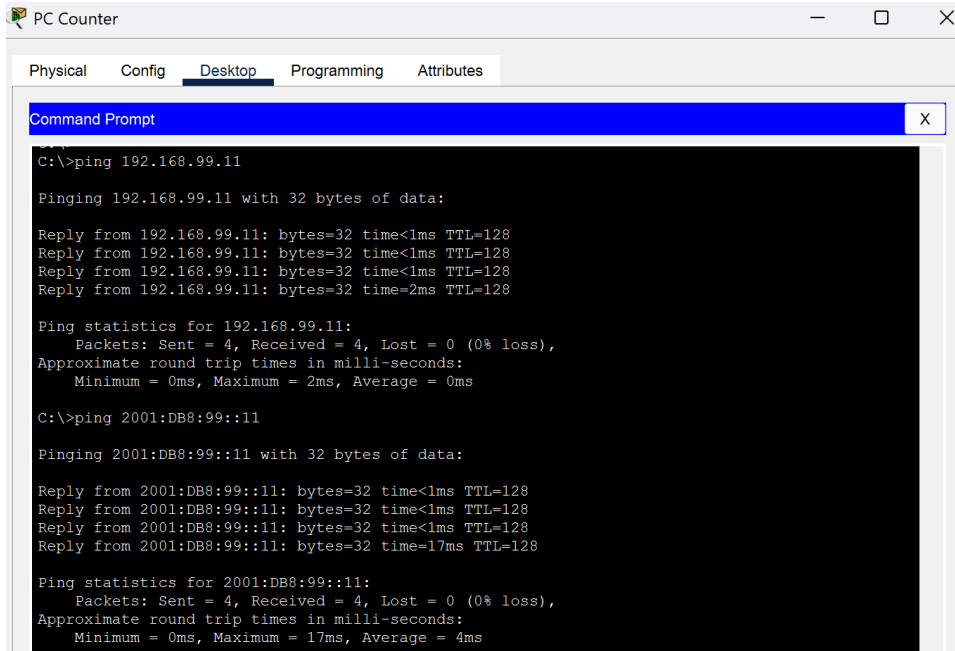
C:\>ping 001:DB8:99::10

Pinging 001:DB8:99::10 with 32 bytes of data:

Reply from 001:DB8:99::10: bytes=32 time<1ms TTL=128
Reply from 001:DB8:99::10: bytes=32 time<1ms TTL=128
Reply from 001:DB8:99::10: bytes=32 time<1ms TTL=128
Reply from 001:DB8:99::10: bytes=32 time<1ms TTL=128

Ping statistics for 001:DB8:99::10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- LAN Server & Admin ke Server DNS



PC Counter

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>ping 192.168.99.11

Pinging 192.168.99.11 with 32 bytes of data:

Reply from 192.168.99.11: bytes=32 time<1ms TTL=128
Reply from 192.168.99.11: bytes=32 time<1ms TTL=128
Reply from 192.168.99.11: bytes=32 time<1ms TTL=128
Reply from 192.168.99.11: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.99.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 2001:DB8:99::11

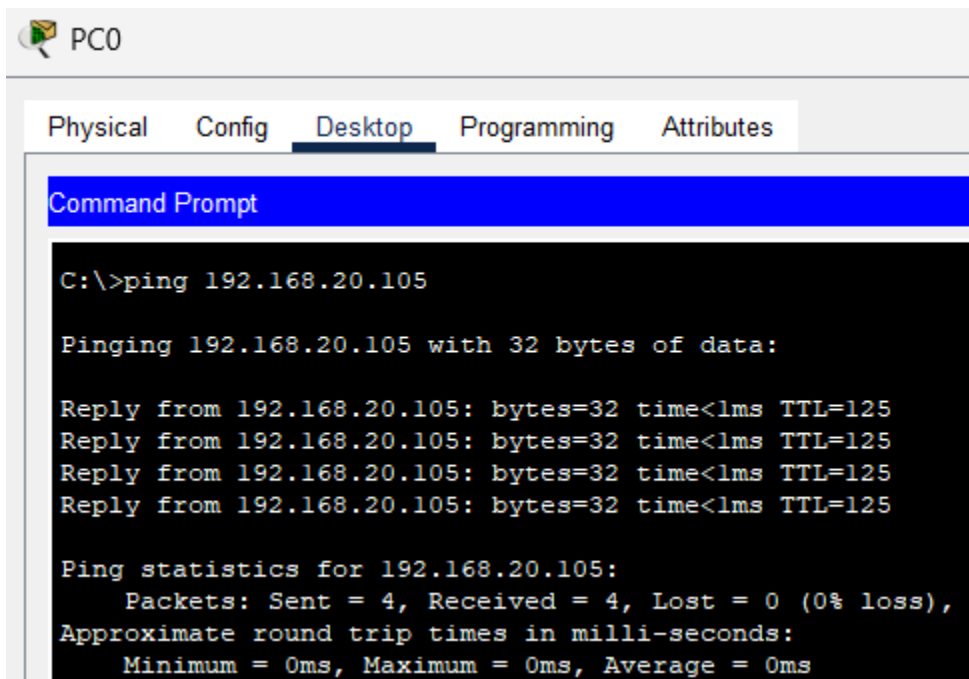
Pinging 2001:DB8:99::11 with 32 bytes of data:

Reply from 2001:DB8:99::11: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:99::11: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:99::11: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:99::11: bytes=32 time=17ms TTL=128

Ping statistics for 2001:DB8:99::11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 4ms
```

Konektivitas Perangkat Regular

- LAN Regular ke LAN VIP (PC0 ke PC17)



PC0

Physical Config **Desktop** Programming Attributes

Command Prompt

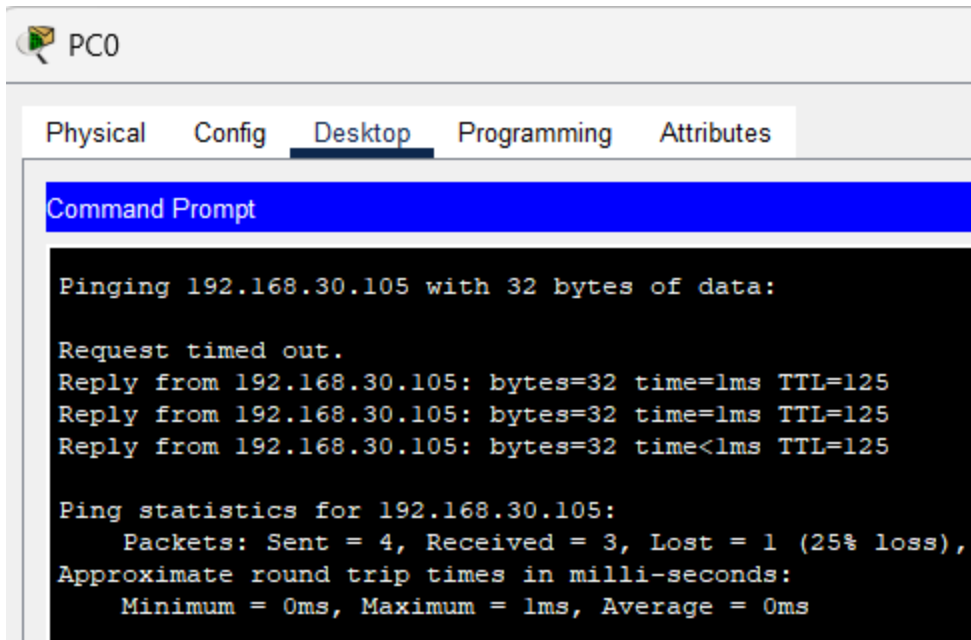
```
C:\>ping 192.168.20.105

Pinging 192.168.20.105 with 32 bytes of data:

Reply from 192.168.20.105: bytes=32 time<1ms TTL=125
Reply from 192.168.20.105: bytes=32 time<1ms TTL=125
Reply from 192.168.20.105: bytes=32 time<1ms TTL=125
Reply from 192.168.20.105: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.20.105:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- LAN Regular ke LAN VVIP (PC 0 ke PC 16)



PC0

Physical Config Desktop Programming Attributes

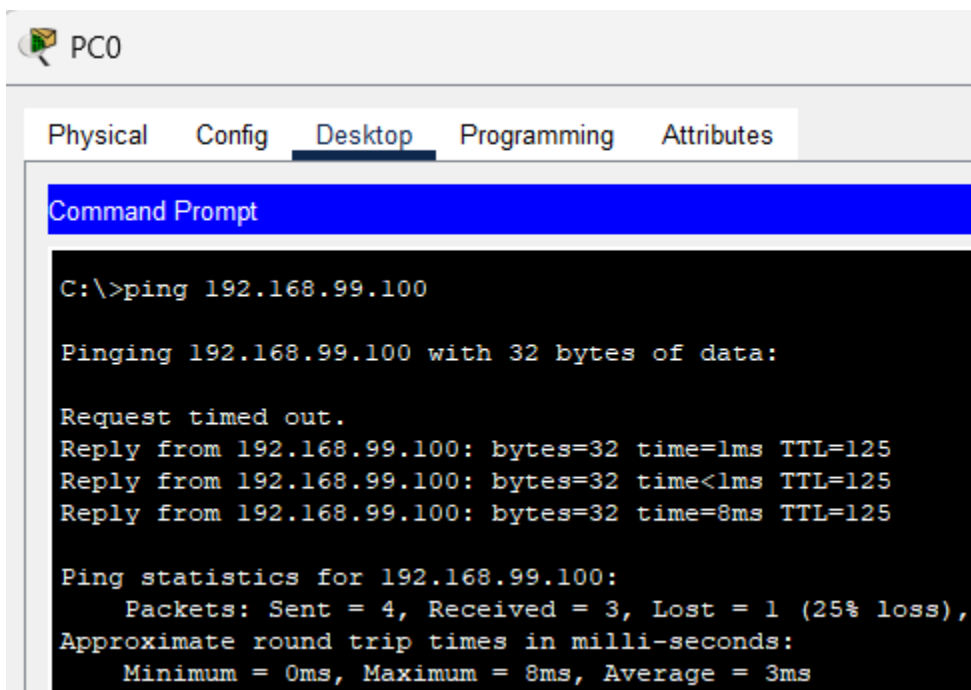
Command Prompt

```
Pinging 192.168.30.105 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.105: bytes=32 time=1ms TTL=125
Reply from 192.168.30.105: bytes=32 time=1ms TTL=125
Reply from 192.168.30.105: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.30.105:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

- LAN Reguler ke LAN Server (PC 0 ke PC counter)



PC0

Physical Config Desktop Programming Attributes

Command Prompt

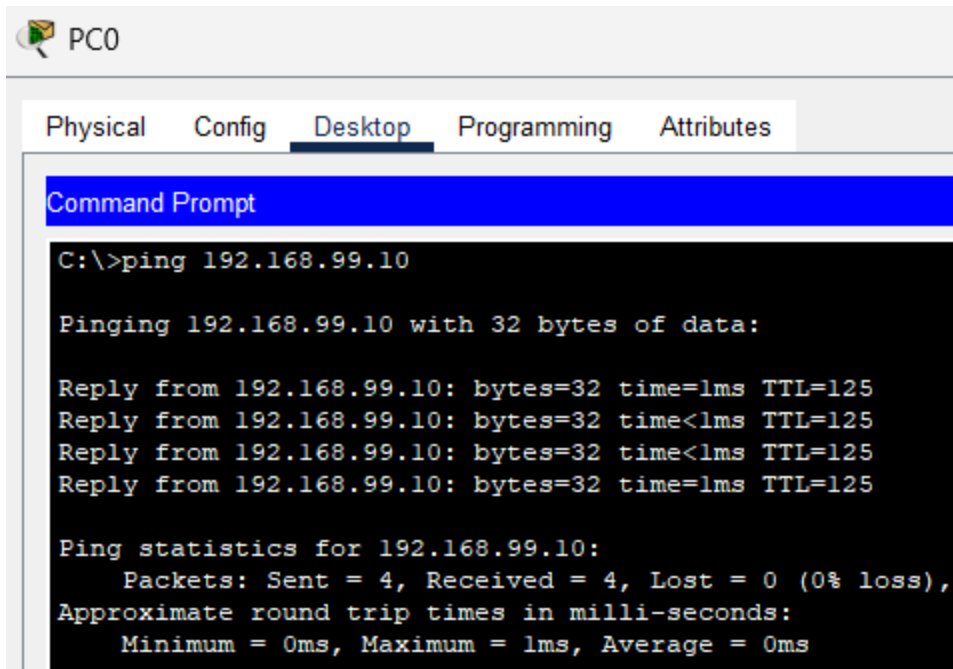
```
C:\>ping 192.168.99.100

Pinging 192.168.99.100 with 32 bytes of data:

Request timed out.
Reply from 192.168.99.100: bytes=32 time=1ms TTL=125
Reply from 192.168.99.100: bytes=32 time<1ms TTL=125
Reply from 192.168.99.100: bytes=32 time=8ms TTL=125

Ping statistics for 192.168.99.100:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 3ms
```

- LAN Reguler ke Server Web



PC0

Physical Config Desktop Programming Attributes

Command Prompt

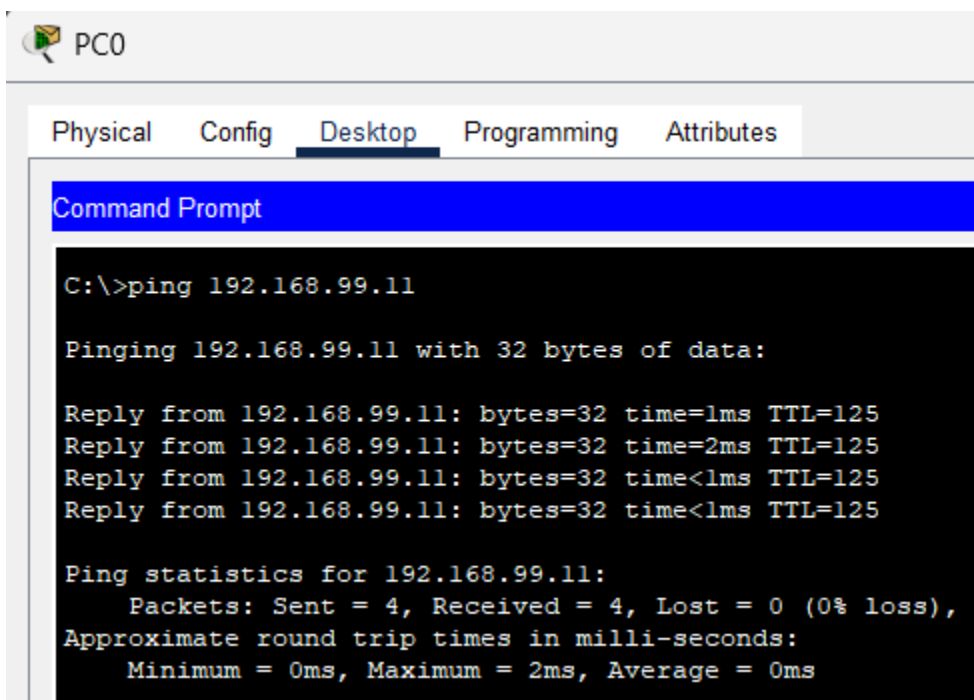
```
C:\>ping 192.168.99.10

Pinging 192.168.99.10 with 32 bytes of data:

Reply from 192.168.99.10: bytes=32 time=1ms TTL=125
Reply from 192.168.99.10: bytes=32 time<1ms TTL=125
Reply from 192.168.99.10: bytes=32 time<1ms TTL=125
Reply from 192.168.99.10: bytes=32 time=1ms TTL=125

Ping statistics for 192.168.99.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

- LAN Reguler ke Server DNS



PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.99.11

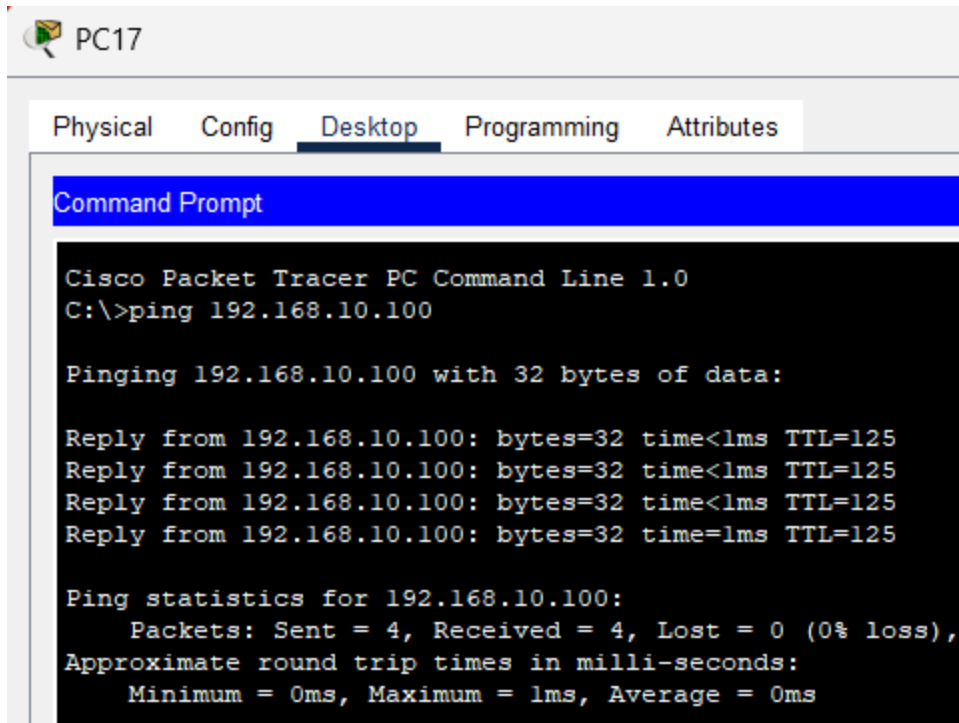
Pinging 192.168.99.11 with 32 bytes of data:

Reply from 192.168.99.11: bytes=32 time=1ms TTL=125
Reply from 192.168.99.11: bytes=32 time=2ms TTL=125
Reply from 192.168.99.11: bytes=32 time<1ms TTL=125
Reply from 192.168.99.11: bytes=32 time<1ms TTL=125

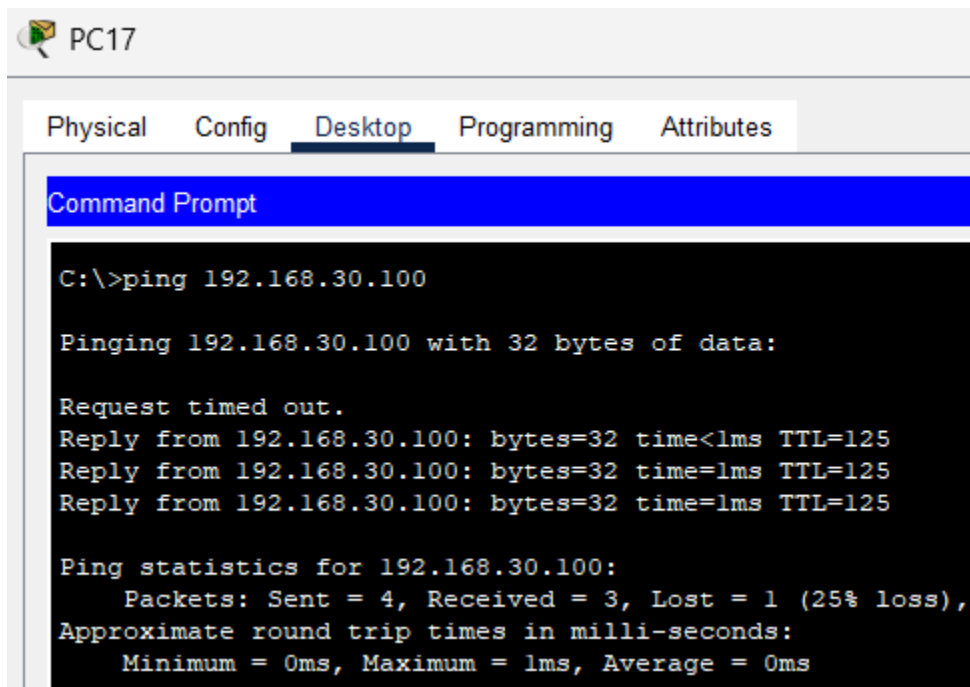
Ping statistics for 192.168.99.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Konektivitas Perangkat VIP

- LAN VIP ke LAN Reguler



- LAN VIP ke LAN VVIP



- LAN VIP ke LAN Server



PC17

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>ping 192.168.99.100

Pinging 192.168.99.100 with 32 bytes of data:

Reply from 192.168.99.100: bytes=32 time=1ms TTL=125
Reply from 192.168.99.100: bytes=32 time=17ms TTL=125
Reply from 192.168.99.100: bytes=32 time<1ms TTL=125
Reply from 192.168.99.100: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.99.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 4ms
```

- LAN Reguler ke Server Web



PC17

Physical Config **Desktop** Programming Attributes

Command Prompt

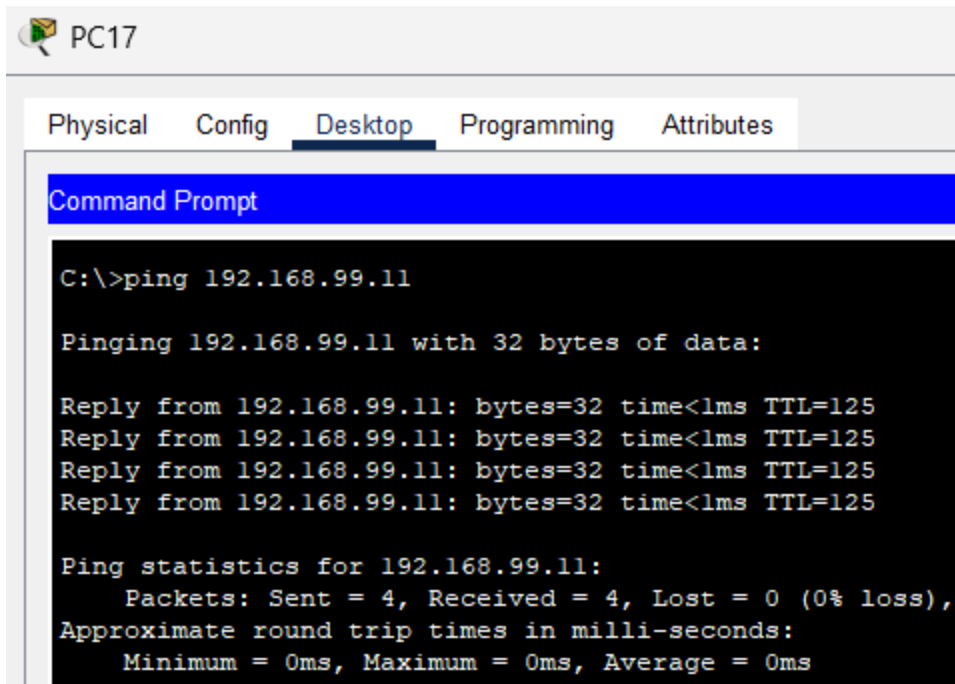
```
C:\>ping 192.168.99.10

Pinging 192.168.99.10 with 32 bytes of data:

Reply from 192.168.99.10: bytes=32 time<1ms TTL=125
Reply from 192.168.99.10: bytes=32 time<1ms TTL=125
Reply from 192.168.99.10: bytes=32 time<1ms TTL=125
Reply from 192.168.99.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.99.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- LAN Reguler ke Server DNS



PC17

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>ping 192.168.99.11

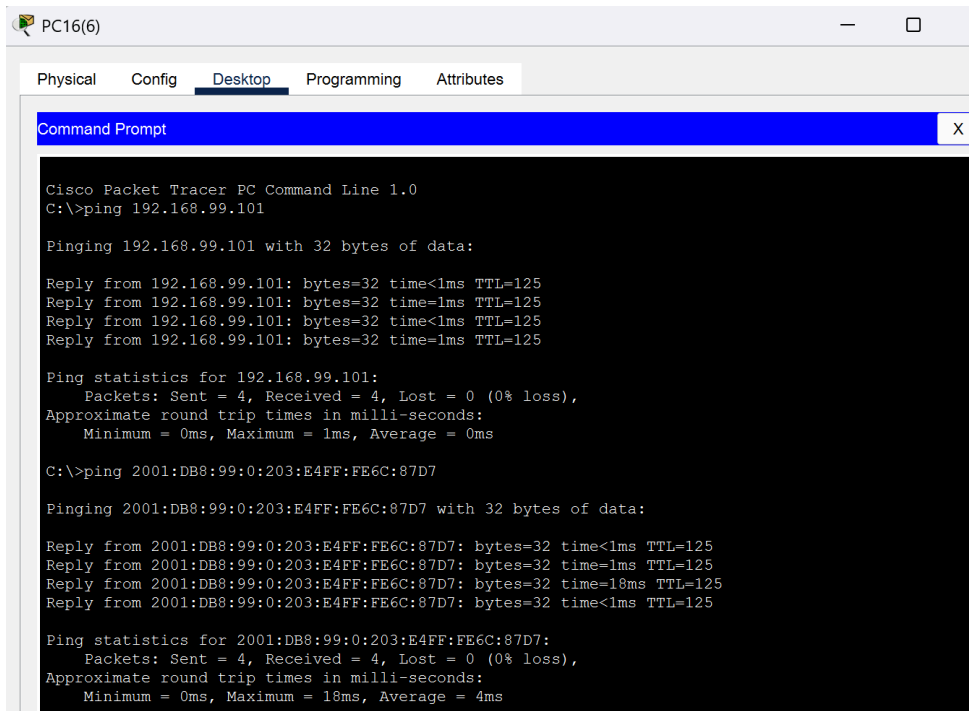
Pinging 192.168.99.11 with 32 bytes of data:

Reply from 192.168.99.11: bytes=32 time<1ms TTL=125
Reply from 192.168.99.11: bytes=32 time<1ms TTL=125
Reply from 192.168.99.11: bytes=32 time<1ms TTL=125
Reply from 192.168.99.11: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.99.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Konektivitas Perangkat VVIP

- LAN VVIP ke LAN Server & Admin (PC 16(6) ke PC Counter)



PC16(6)

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.99.101

Pinging 192.168.99.101 with 32 bytes of data:

Reply from 192.168.99.101: bytes=32 time<1ms TTL=125
Reply from 192.168.99.101: bytes=32 time<1ms TTL=125
Reply from 192.168.99.101: bytes=32 time<1ms TTL=125
Reply from 192.168.99.101: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.99.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 2001:DB8:99:0:203:E4FF:FE6C:87D7

Pinging 2001:DB8:99:0:203:E4FF:FE6C:87D7 with 32 bytes of data:

Reply from 2001:DB8:99:0:203:E4FF:FE6C:87D7: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:99:0:203:E4FF:FE6C:87D7: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:99:0:203:E4FF:FE6C:87D7: bytes=32 time=18ms TTL=125
Reply from 2001:DB8:99:0:203:E4FF:FE6C:87D7: bytes=32 time<1ms TTL=125

Ping statistics for 2001:DB8:99:0:203:E4FF:FE6C:87D7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 4ms
```

- LAN VVIP ke LAN Regular (PC16(6) ke PC14)

PC16(6)

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>ping 192.168.10.102

Pinging 192.168.10.102 with 32 bytes of data:

Reply from 192.168.10.102: bytes=32 time<1ms TTL=125
Reply from 192.168.10.102: bytes=32 time<1ms TTL=125
Reply from 192.168.10.102: bytes=32 time<1ms TTL=125
Reply from 192.168.10.102: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.10.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:DB8:10:0:201:42FF:FE79:774E

Pinging 2001:DB8:10:0:201:42FF:FE79:774E with 32 bytes of data:

Reply from 2001:DB8:10:0:201:42FF:FE79:774E: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:10:0:201:42FF:FE79:774E: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:10:0:201:42FF:FE79:774E: bytes=32 time=2ms TTL=125
Reply from 2001:DB8:10:0:201:42FF:FE79:774E: bytes=32 time<1ms TTL=125

Ping statistics for 2001:DB8:10:0:201:42FF:FE79:774E:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

- LAN VVIP ke LAN VIP

PC16(6)

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>ping 192.168.20.102

Pinging 192.168.20.102 with 32 bytes of data:

Reply from 192.168.20.102: bytes=32 time<1ms TTL=125
Reply from 192.168.20.102: bytes=32 time<1ms TTL=125
Reply from 192.168.20.102: bytes=32 time<1ms TTL=125
Reply from 192.168.20.102: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.20.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

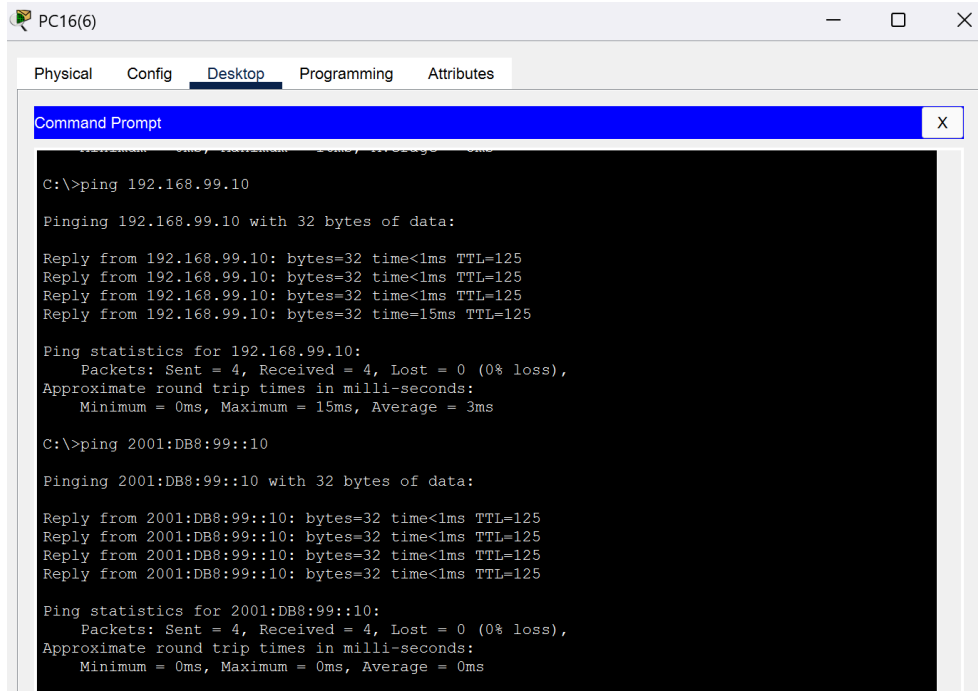
C:\>ping 2001:DB8:20:0:201:96FF:FEB6:9AB4

Pinging 2001:DB8:20:0:201:96FF:FEB6:9AB4 with 32 bytes of data:

Reply from 2001:DB8:20:0:201:96FF:FEB6:9AB4: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:20:0:201:96FF:FEB6:9AB4: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:20:0:201:96FF:FEB6:9AB4: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:20:0:201:96FF:FEB6:9AB4: bytes=32 time<1ms TTL=125

Ping statistics for 2001:DB8:20:0:201:96FF:FEB6:9AB4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- LAN VVIP ke Server Web



PC16(6)

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.99.10

Pinging 192.168.99.10 with 32 bytes of data:

Reply from 192.168.99.10: bytes=32 time<1ms TTL=125
Reply from 192.168.99.10: bytes=32 time<1ms TTL=125
Reply from 192.168.99.10: bytes=32 time<1ms TTL=125
Reply from 192.168.99.10: bytes=32 time=15ms TTL=125

Ping statistics for 192.168.99.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 3ms

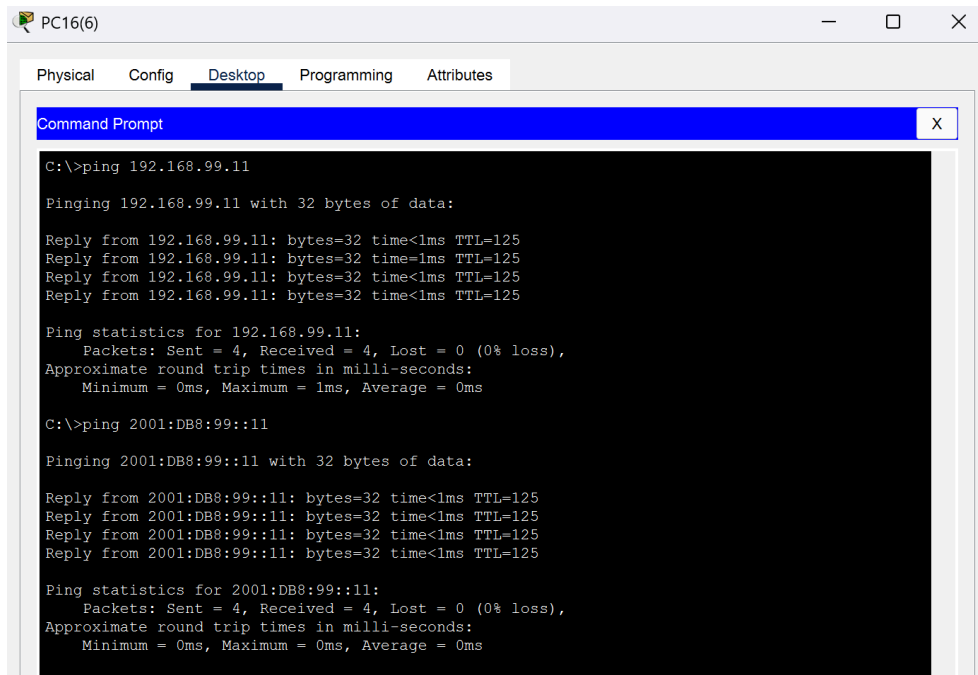
C:\>ping 2001:DB8:99::10

Pinging 2001:DB8:99::10 with 32 bytes of data:

Reply from 2001:DB8:99::10: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:99::10: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:99::10: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:99::10: bytes=32 time<1ms TTL=125

Ping statistics for 2001:DB8:99::10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- LAN VVIP ke Server DNS



PC16(6)

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.99.11

Pinging 192.168.99.11 with 32 bytes of data:

Reply from 192.168.99.11: bytes=32 time<1ms TTL=125
Reply from 192.168.99.11: bytes=32 time=1ms TTL=125
Reply from 192.168.99.11: bytes=32 time<1ms TTL=125
Reply from 192.168.99.11: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.99.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 2001:DB8:99::11

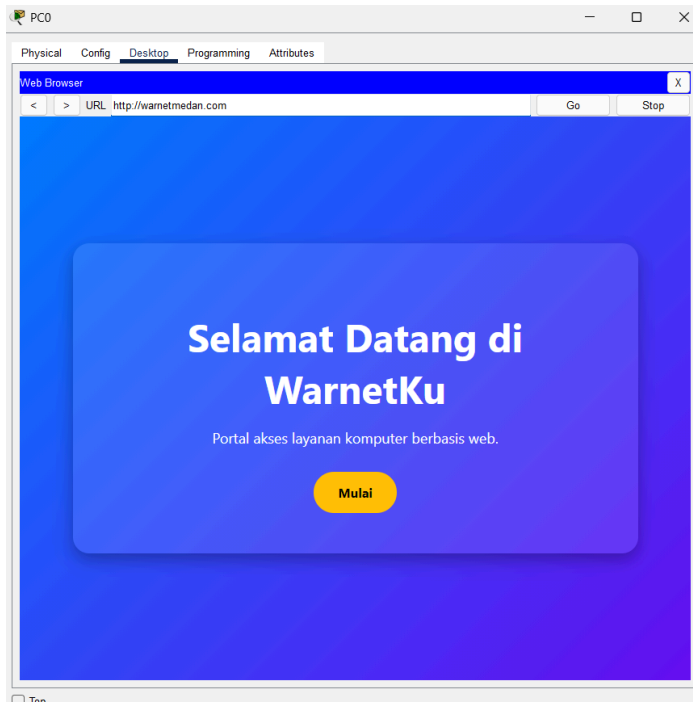
Pinging 2001:DB8:99::11 with 32 bytes of data:

Reply from 2001:DB8:99::11: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:99::11: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:99::11: bytes=32 time<1ms TTL=125
Reply from 2001:DB8:99::11: bytes=32 time<1ms TTL=125

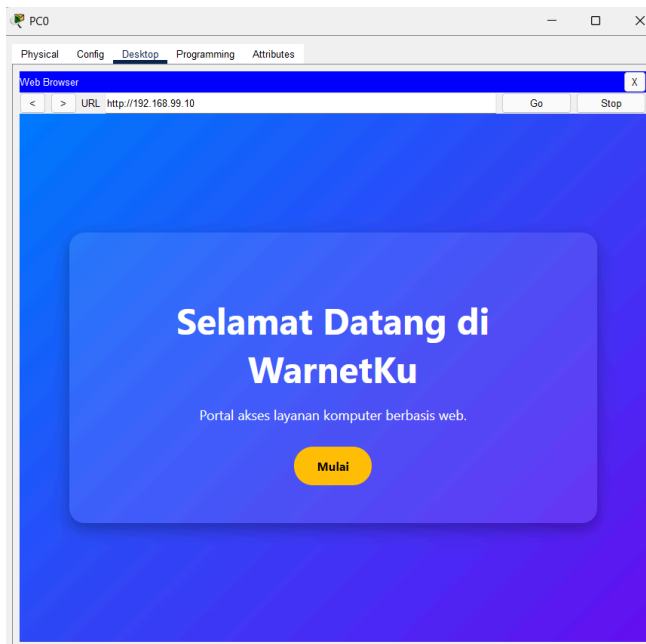
Ping statistics for 2001:DB8:99::11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Tampilan Web

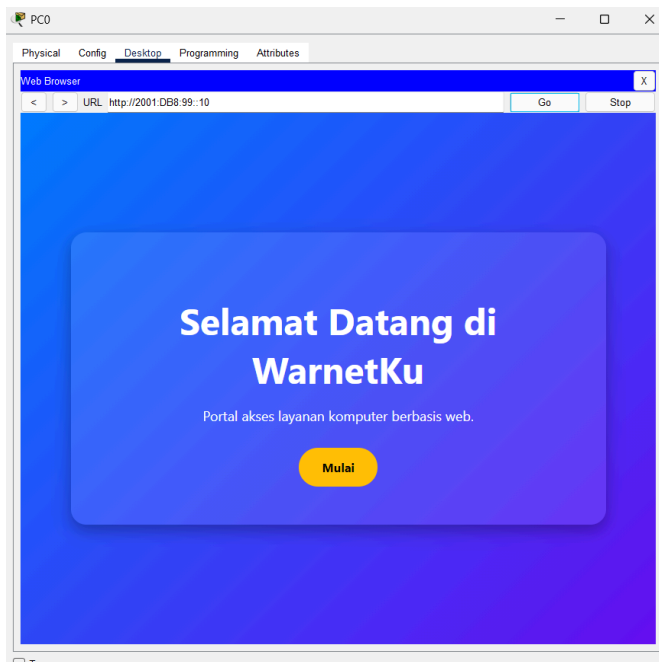
- Jika akses dengan nama web dan domain



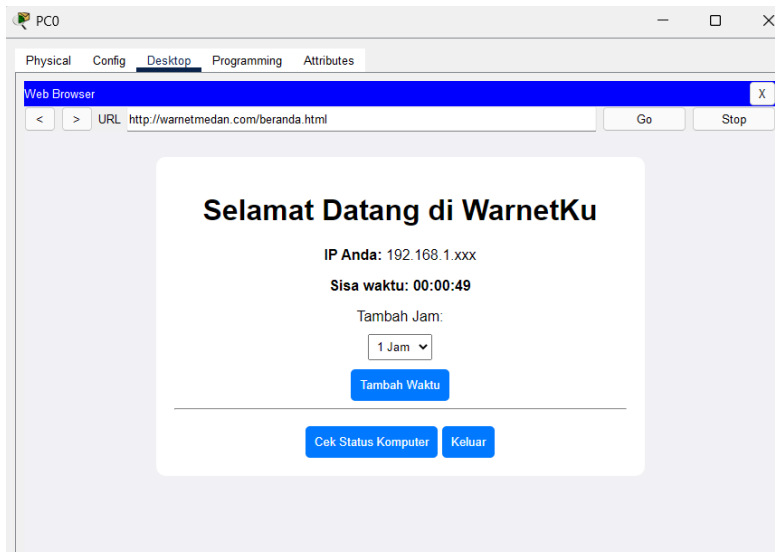
- Jika Akses dengan IPv4



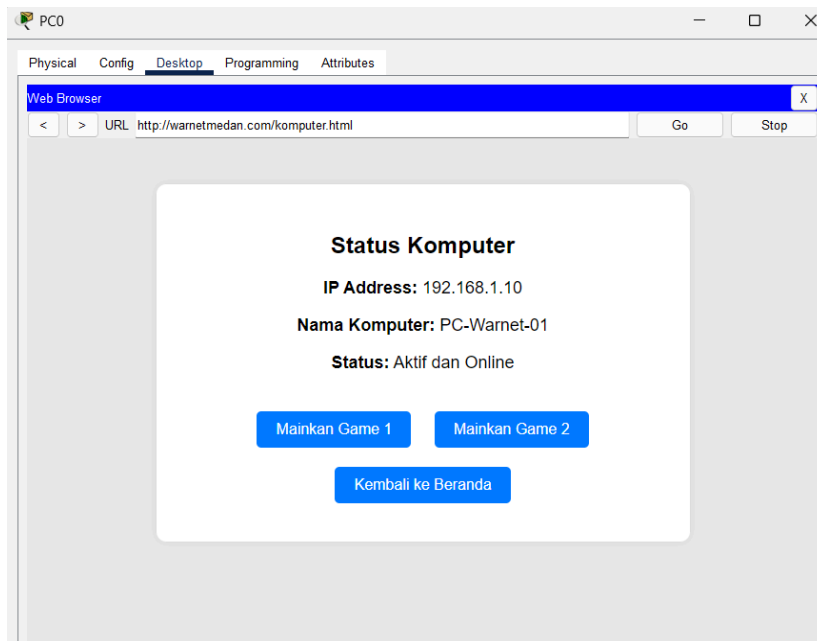
- Jika akses dengan IPv6



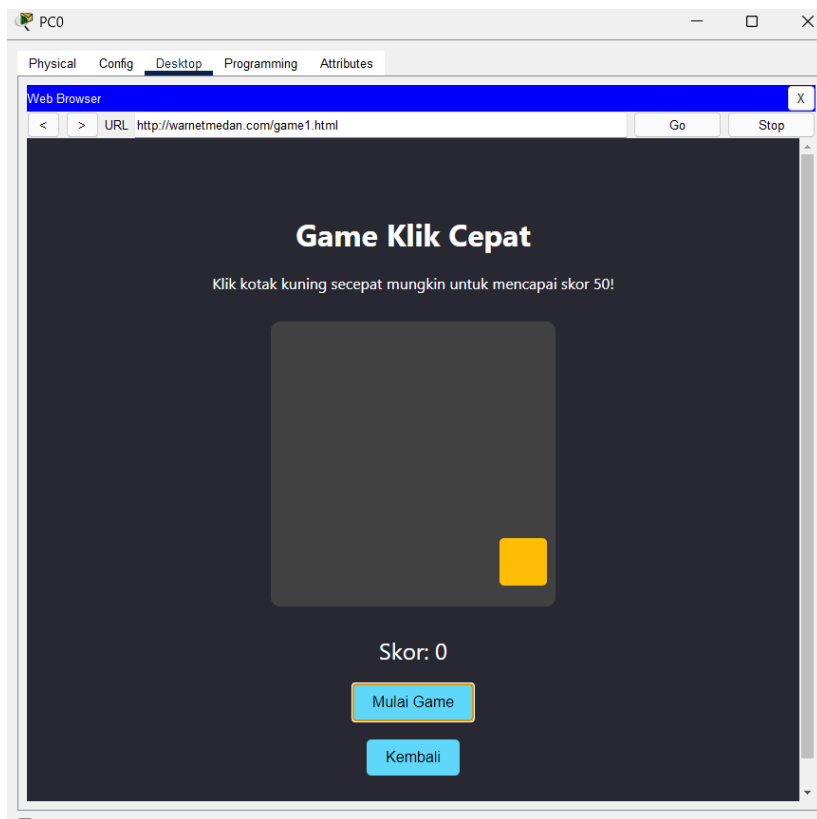
- Beranda



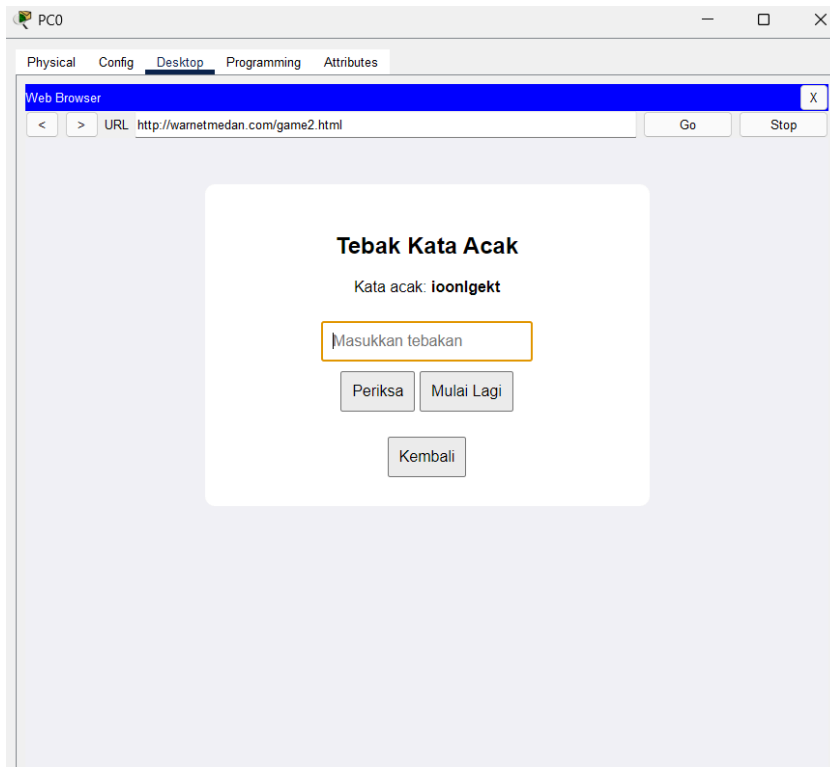
- Status Komputer



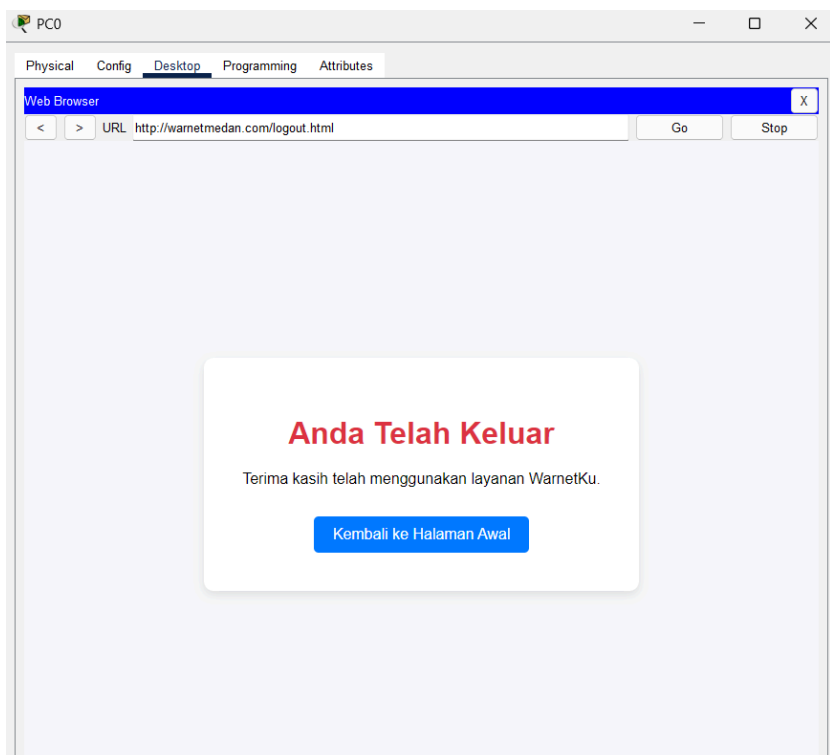
- Game 1



- Game 2



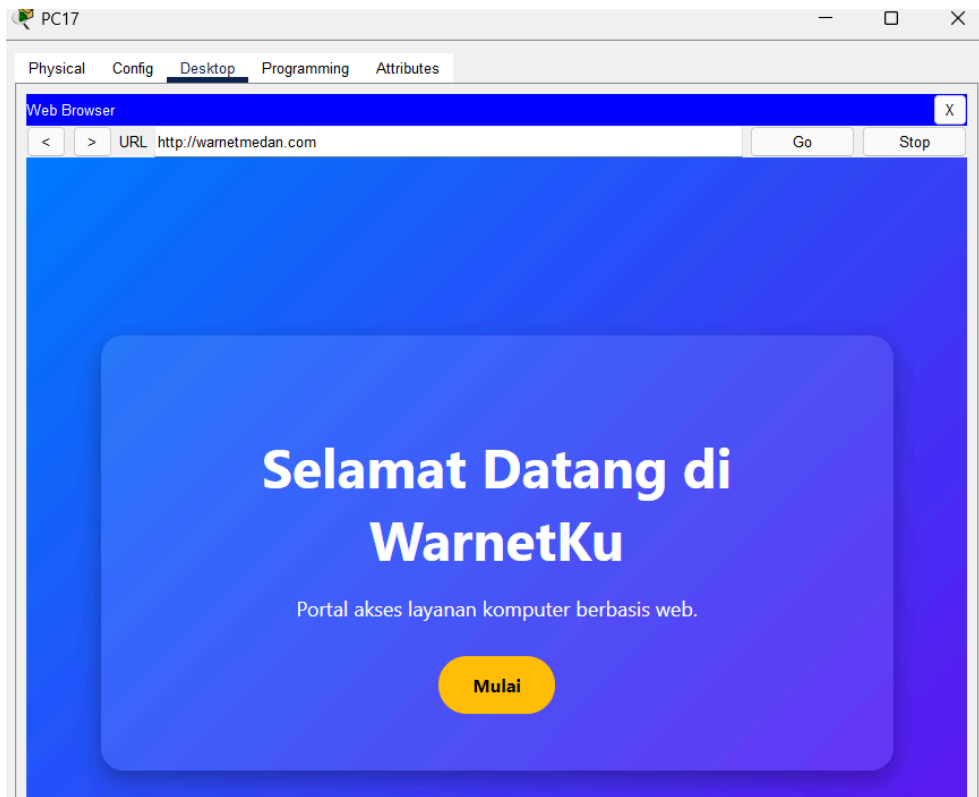
- Logout



10. Tulis Hasil Eksplorasi yang Dilakukan

- Server Web dan Server DNS

Server Web dan DNS digunakan untuk web dari Warnet dan Domain nya



- VLAN

VLAN digunakan untuk pemberian IP pada Interface FastEthernet pada Router Internet karena menggunakan modul HSIW-4ESW yang tidak dapat langsung konfigurasi IP nya (VLAN data)


```
Internet
Physical Config CLI Attributes

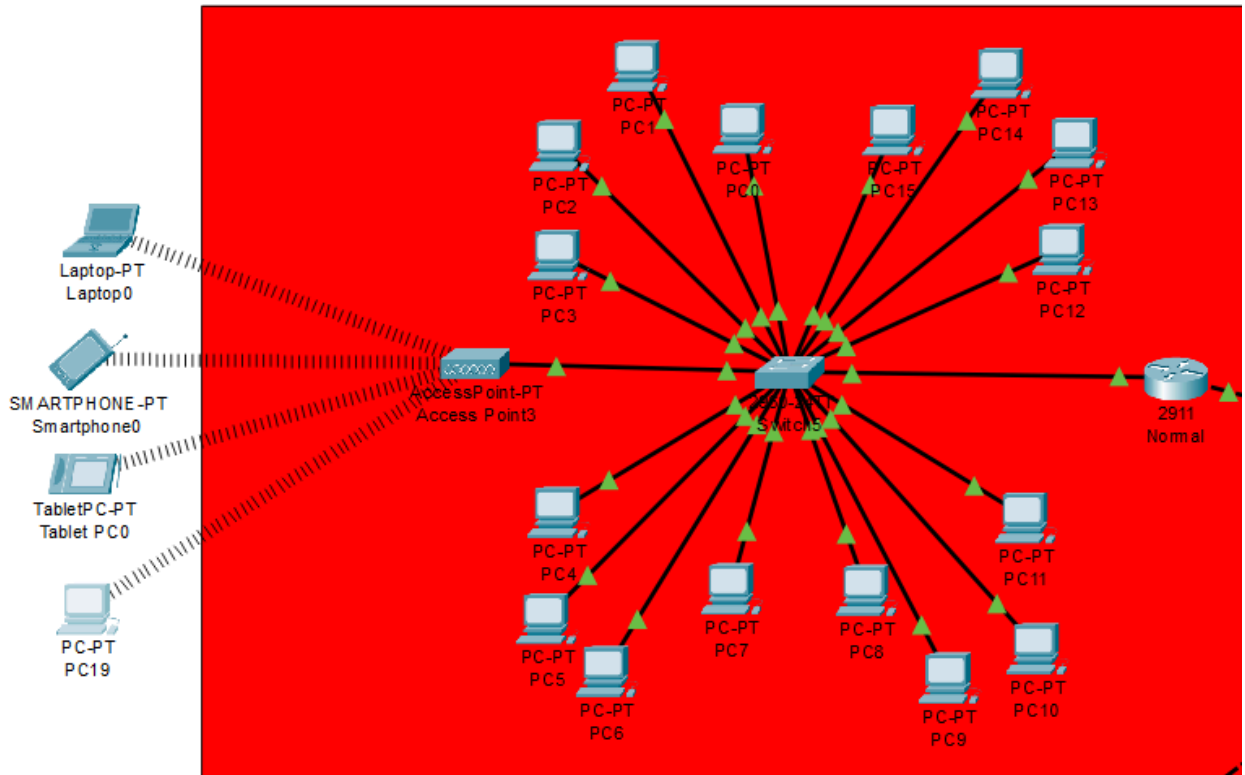
!
interface Vlan10
description Link to R1
mac-address 00e0.b090.2001
ip address 10.0.1.1 255.255.255.252
ip nat inside
ipv6 address 2001:DB8:1::1/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface Vlan20
description Link to R2
mac-address 00e0.b090.2002
ip address 10.0.2.1 255.255.255.252
ip nat inside
ipv6 address 2001:DB8:2::1/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface Vlan30
description Link to R3
mac-address 00e0.b090.2003
ip address 10.0.3.1 255.255.255.252
ip nat inside
ipv6 address 2001:DB8:3::1/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface Vlan40
description Link to R4
mac-address 00e0.b090.2004
ip address 10.0.4.1 255.255.255.252
ip nat inside
ipv6 address 2001:DB8:4::1/64
ipv6 enable
ipv6 ospf 1 area 0
!
```

```
Internet
Physical Config CLI Attribute

!
interface FastEthernet0/3/0
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/3/1
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/3/2
switchport access vlan 30
switchport mode access
!
interface FastEthernet0/3/3
switchport access vlan 40
switchport mode access
```

- WLAN (contoh di R1)

WLAN digunakan untuk koneksi perangkat ke LAN secara wireless, kami menggunakan Access Point untuk WLAN



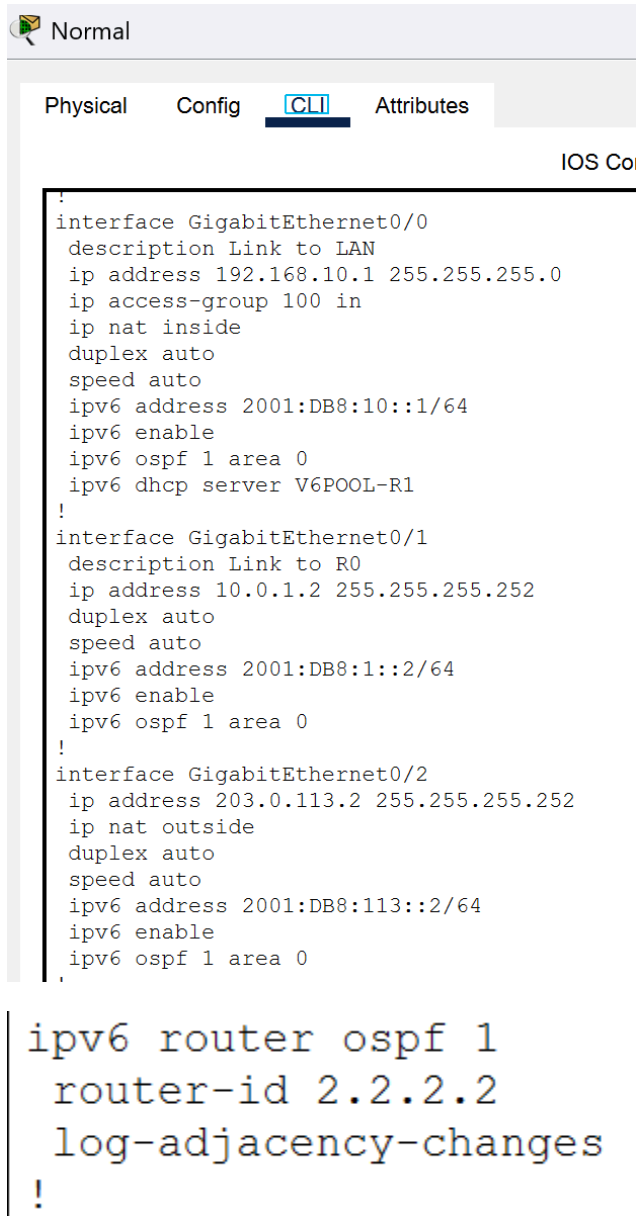
- Subnetting IPV6 (IPV6 R1 g0/0)

Subnetting IPV6 digunakan untuk membagi jaringan besar menjadi jaringan-jaringan kecil, untuk mengidentifikasi subnet untuk LAN yang terhubung, serta memisahkan jaringan agar lebih aman dan teratur

```
!
interface GigabitEthernet0/0
description Link to LAN
ip address 192.168.10.1 255.255.255.0
ip access-group 100 in
ip nat inside
duplex auto
speed auto
ipv6 address 2001:DB8:10::1/64
```

- Routing IPV6 (R1)

Routing IPV6 digunakan untuk menghubungkan antar subnet IPV6, memungkinkan komunikasi antar router. Untuk routing IPV6 harus membuat ospf di setiap interface yang ada di dalam router, kemudian membuat router-id.



```
!
interface GigabitEthernet0/0
  description Link to LAN
  ip address 192.168.10.1 255.255.255.0
  ip access-group 100 in
  ip nat inside
  duplex auto
  speed auto
  ipv6 address 2001:DB8:10::1/64
  ipv6 enable
  ipv6 ospf 1 area 0
  ipv6 dhcp server V6POOL-R1
!
interface GigabitEthernet0/1
  description Link to R0
  ip address 10.0.1.2 255.255.255.252
  duplex auto
  speed auto
  ipv6 address 2001:DB8:1::2/64
  ipv6 enable
  ipv6 ospf 1 area 0
!
interface GigabitEthernet0/2
  ip address 203.0.113.2 255.255.255.252
  ip nat outside
  duplex auto
  speed auto
  ipv6 address 2001:DB8:113::2/64
  ipv6 enable
  ipv6 ospf 1 area 0
!

ipv6 router ospf 1
  router-id 2.2.2.2
  log-adjacency-changes
!
```

- SSH

SSH digunakan untuk membatasi akses agar PC user agar tidak dapat mengakses router. Untuk membuat SSH di router kita perlu membuat domain-name, kemudian username untuk admin maupun user beserta passwordnya kemudian di encrypt menggunakan 'crypto key generate rsa' dan masukkan rekomendasi panjang bit key 1024. Kemudian membuat akses list ip mana saja yang memiliki izin untuk mengakses dan yang tidak dapat mengakses.

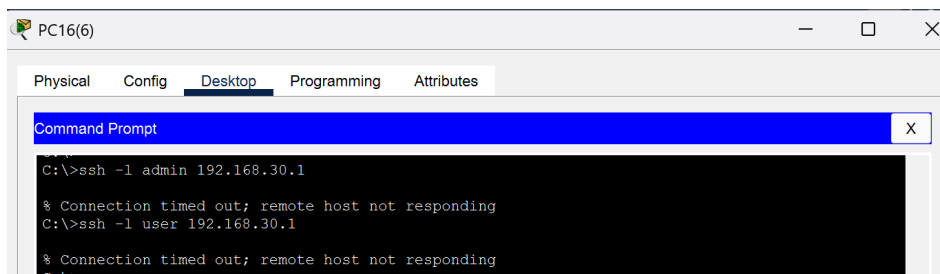
```
ip domain-name belajar.com
```

```
username admin privilege 15 secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/  
username user secret 5 $1$mERr$pRhYIeRsLJ9jhTe0lpBTz/
```

```
line vty 0 4  
login local  
transport input ssh
```

```
access-list 100 permit tcp host 192.168.99.100 any eq 22  
access-list 100 permit tcp host 192.168.99.101 any eq 22  
access-list 100 deny tcp any any eq 22  
access-list 100 permit ip any any
```

Tes SSH dari PC user VVIP ke R3



Tes SSH dari PC Server & Admin ke R3. Jika username yang digunakan admin maka admin dapat melakukan perubahan pada router namun jika menggunakan username user maka hanya dapat masuk tetapi tidak dapat melakukan enable terhadap router.

